

TALLER 6

Notación de índices

Escalar: ϕ – simbólico

Componente vector: $A = [A_1 \ A_2 \ A_3]$ – simbólico

Tensor simbólico: $T = T_{ij} e_i e_j$ – todos los índices se ponen

Convención de suma de Einstein

Índices repetidos = se suman (convención de Einstein). Índice libre.

Casos especiales

Son términos formas.

$$\left(\frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} \right) dx_i dx_j = \text{forma cuadrática}$$

$$\frac{\partial(T_{ij} x_j)}{\partial x_k} = T_{ik}$$

$$\frac{\partial(x_i x_j)}{\partial x_k} = \delta_{ik} x_j + x_i \delta_{jk}$$

En coordenadas cartesianas: $\vec{r} = e_i x_i = (x, y, z)$. En coordenadas curvilíneas:

$$d\vec{r} = dx_i e_i + x_i de_i$$

$$ds^2 = dx_i dx_i = g_{ij} dx_i dx_j$$

Traza:

$$\text{Tr } S = S_{11} + S_{22} + S_{33} = S_{ii}$$

$$S_{ij} S_{ij} = S_{11}^2 + S_{12}^2 + S_{13}^2 + S_{21}^2 + S_{22}^2 + S_{23}^2 + S_{31}^2 + S_{32}^2 + S_{33}^2$$

$$S_{ij} S_{ji} = S_{11}^2 + S_{12}S_{21} + S_{13}S_{31} + S_{21}S_{12} + S_{22}^2 + S_{23}S_{32} + S_{31}S_{13} + S_{32}S_{23} + S_{33}^2$$

$$S_{jk} S_{kj} = S_{11}^2 + S_{12}S_{21} + S_{13}S_{31} + S_{21}S_{12} + S_{22}^2 + S_{23}S_{32} + S_{31}S_{13} + S_{32}S_{23} + S_{33}^2$$

$$S_{ij} S_{ij} = S_{11}^2 + S_{12}^2 + S_{13}^2 + S_{21}^2 + S_{22}^2 + S_{23}^2 + S_{31}^2 + S_{32}^2 + S_{33}^2$$

$$S_{mn} a_m a_n = S_{11}a_1^2 + S_{12}a_1a_2 + S_{13}a_1a_3 + S_{21}a_2a_1 + S_{22}a_2^2 + S_{23}a_2a_3 + S_{31}a_3a_1 + S_{32}a_3a_2 + S_{33}a_3^2$$

Tensor simétrico / antisimétrico:

Si S es simétrico: $S_{ij} = S_{ji}$

Si A es antisimétrico: $A_{ij} = -A_{ji}$

$$A_{ii} = 0 \quad (\text{diagonal nula para antisimétrico})$$

$$S_{ij} A_{ij} = 0 \quad (\text{si } S \text{ simétrico y } A \text{ antisimétrico})$$

Determinante:

$$\det(A) = \varepsilon_{ijk} A_{1i} A_{2j} A_{3k}$$

$$= A_{11}(A_{22}A_{33} - A_{23}A_{32}) - A_{12}(A_{21}A_{33} - A_{23}A_{31}) + A_{13}(A_{21}A_{32} - A_{22}A_{31})$$

Producto cruz:

$$(\vec{A} \times \vec{B})_i = \varepsilon_{ijk} A_j B_k$$

$$(\vec{A} \times \vec{B}) \cdot \vec{C} = \varepsilon_{ijk} A_i B_j C_k$$

$$(\vec{A} \times \vec{B}) \cdot (\vec{A} \times \vec{B}) = A^2 B^2 - (\vec{A} \cdot \vec{B})^2$$

Transformaciones de coordenadas:

$r'(T)$, $w'(T)$ invariancia. Una transformación de coordenadas dada por:

$$T'_{ij} = a_{ik} a_{jl} T_{kl}$$

Transformaciones de coordenadas:

$$\text{Tr}(T') = \text{Tr}(T)$$

$$a_{ik} a_{jk} = \delta_{ij}, \quad a_{ki} a_{kj} = \delta_{ij}$$

$$\text{Tr}(T') = a_{ik} a_{jl} T_{kl} \delta_{ij} = a_{ik} a_{il} T_{kl} = \delta_{kl} T_{kl} = T_{kk} = \text{Tr}(T)$$

$$\det(T') = \det(A) \det(T) \det(A^T) = \det(T) (\det A)^2 = \det(T)$$

$$\det(A) = \pm 1$$

Identidades tensoriales:

$$\nabla \cdot (\vec{A} \vec{B}) = \vec{B} (\nabla \cdot \vec{A}) + (\vec{A} \cdot \nabla) \vec{B}$$

$$\nabla (\vec{A} \cdot \vec{B}) = (\nabla \vec{A}) \cdot \vec{B} + (\nabla \vec{B}) \cdot \vec{A}$$

$$\nabla \times (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \vec{A}) - \vec{A} \cdot (\nabla \vec{B}) + \vec{A} (\nabla \cdot \vec{B}) - \vec{B} (\nabla \cdot \vec{A})$$

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$$

$$\nabla \cdot (\mathbf{T} \times \vec{A}) = (\nabla \cdot \mathbf{T}) \times \vec{A} + \mathbf{T} : (\nabla \times \vec{A})$$

$$\nabla \cdot (\mathbf{T} \cdot \vec{A}) = (\nabla \cdot \mathbf{T}) \cdot \vec{A} + \mathbf{T} : (\nabla \vec{A})$$

$$\nabla (\mathbf{T} : \mathbf{R}) = (\nabla \mathbf{T}) : \mathbf{R} + (\nabla \mathbf{R}) : \mathbf{T}$$

$$\nabla \cdot (\mathbf{T} \cdot \mathbf{R}) = (\nabla \cdot \mathbf{T}) \cdot \mathbf{R} + \mathbf{T}^T : (\nabla \mathbf{R})$$

$$\nabla (\vec{A} \cdot \mathbf{T} \cdot \vec{B}) = (\nabla \vec{A}) \cdot \mathbf{T} \cdot \vec{B} + (\nabla \mathbf{T}) : \vec{A} \vec{B} + (\nabla \vec{B}) \cdot \mathbf{T}^T \cdot \vec{A}$$